

- 2 Fig. 2 shows two complex numbers z_1 and z_2 represented on an Argand diagram.

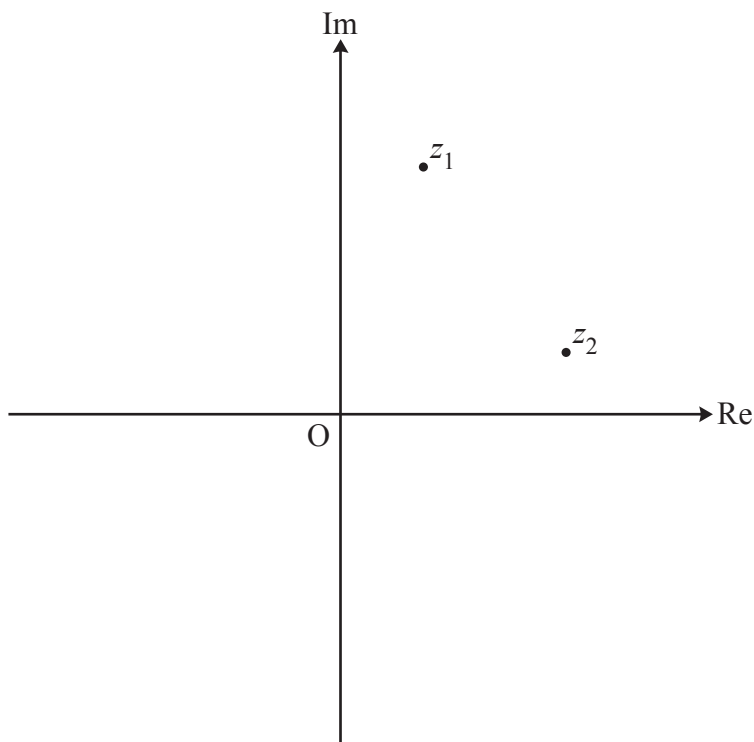


Fig. 2

- (a) On the copy of Fig. 2 in the Printed Answer Booklet, mark points representing each of the following complex numbers.

- z_1^*

- $z_2 - z_1$

[2]

- (b) In this question you must show detailed reasoning.

In the case where $z_1 = 1 + 2i$ and $z_2 = 3 + i$, find $\frac{z_2 - z_1}{z_1^*}$ in the form $a + ib$, where a and b are real numbers.

[2]